

A Comparative Review of Machine Learning and Deep Learning Models for Stock Market Price and Trend Prediction

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Abstract

Purpose: This review systematically compares machine learning (ML) and deep learning (DL) models for stock market forecasting, focusing on their effectiveness when incorporating technical indicators.

Methodology: We analyze Support Vector Machines (SVM), Random Forests, XGBoost, Long Short-Term Memory (LSTM) networks, CNN-LSTM hybrids, and Transformers across multiple performance metrics and market contexts through systematic literature review.

Findings: Deep learning models (LSTM, CNN-LSTM, and Transformers) show improved performance over traditional ML approaches in capturing temporal dependencies. Recent studies report LSTM achieving MAPE values of 2.72–16.21%, CNN-LSTM models demonstrating 15% RMSE improvements over standalone architectures, and Temporal Fusion Transformers achieving MAPE below 2% in specific contexts, though performance varies significantly across markets and time periods.

Implications: For practitioners, model selection should balance predictive performance with computational costs and interpretability requirements. Hybrid architectures offer promising directions but require careful validation across diverse market conditions.

Originality: This paper provides an updated assessment of model architectures and technical indicators based on recent peer-reviewed research, offering evidence-based insights for researchers and practitioners in the financial domain.

Keywords: stock market prediction, machine learning, deep learning, technical indicators, LSTM, CNN-LSTM, Transformers, SVM, Random Forests, XGBoost

1 Introduction

Stock market prediction remains a central challenge in computational finance, addressing price volatility driven by economic, geopolitical, and behavioral factors. Traditional approaches, such as fundamental and technical analysis, often struggle with the non-linear and stochastic nature of markets, prompting the adoption of advanced computational methods like machine learning (ML) and deep learning (DL). ML models, including Support Vector Machines (SVM), Random Forests, and XGBoost, leverage statistical patterns to forecast trends. In contrast, DL models, such as Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Transformers, excel in modeling complex temporal and sequential relationships inherent in time series data.

Technical indicators, such as Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and Bollinger Bands, play a pivotal role by quantifying market momentum and trends. This review aims to:

- 1) Compare ML and DL models for stock price and trend prediction using established performance metrics.
- 2) Evaluate the role of technical indicators in improving forecasting accuracy.
- 3) Identify current best practices and realistic performance expectations.
- 4) Provide evidence-based recommendations for model selection.

By critically assessing recent peer-reviewed research, this paper seeks to guide researchers and practitioners toward effective forecasting solutions with realistic expectations about model capabilities and limitations.

2 Literature Review

The literature on stock market prediction reflects a dynamic shift toward ML and DL approaches, driven by their ability to model complex patterns in historical price data. This section synthesizes recent studies from high-quality journals and conference proceedings, highlighting methodologies, performance metrics, and research gaps.

2.1 Traditional Statistical Models

Early research emphasized statistical models like Autoregressive Integrated Moving Average (ARIMA), which offer robust short-term forecasting capabilities but demonstrate limitations in adapting to market regime changes and exogenous shocks. Recent comparative studies show ARIMA achieving accuracy rates of 89.8% for short-term S&P 500 predictions, but with declining performance over longer horizons [7].

2.2 Machine Learning Approaches

Traditional ML models have demonstrated moderate success in stock prediction tasks. Support Vector Machines (SVM) remain popular for classification tasks, achieving directional prediction accuracies in the 55–65% range for realistic market conditions [1]. Ensemble methods like Random Forests and XGBoost show improved performance through feature importance ranking and non-linear pattern recognition, though they require extensive feature engineering and struggle with purely temporal dependencies.

2.3 Deep Learning Innovations

Deep learning models have gained prominence for their ability to model temporal dependencies without extensive manual feature engineering:

LSTM Networks: Recent research demonstrates LSTM achieving variable performance across different stocks and timeframes. A comprehensive study on NASDAQ stocks reported MAPE values ranging from 2.72% (Apple) to 8.05% (AAL) on test data, with the model capturing both short and long-term dependencies [8, 11]. However, performance is highly sensitive to market volatility and training data quality.

CNN-LSTM Hybrids: Hybrid architectures combining CNN’s spatial feature extraction with LSTM’s temporal modeling show promising results. A study on Indian NSE stocks reported 15% RMSE improvement over standalone models, with particular effectiveness during volatile market conditions [6]. Another implementation achieved RMSE of 0.0064 (training) and 0.03 (testing) using genetic algorithm optimization [4]. The widely-cited Lu et al. study on CNN-LSTM demonstrated effective feature extraction from historical stock data across 7,127 trading days [2].

Transformers: Temporal Fusion Transformers (TFT) represent the current state-of-the-art for multi-horizon forecasting. Recent implementations report:

- MAPE of 1.99% on Apple stock data (2015–2025) when integrating visual time-series representations [9]
- 40–50% reduction in MAE compared to LSTM/BiLSTM on Vietnamese stocks (VIC, VRE, VHM), achieving MAPE below 2% [10]
- Superior performance in capturing long-range dependencies through self-attention mechanisms [3]

However, these results should be interpreted with caution as they represent optimal conditions on specific datasets and may not generalize across all market contexts.

2.4 Research Gaps and Limitations

Despite advances, several challenges persist:

- **Performance Variability:** Model performance varies significantly across different stocks, time periods, and market conditions, with reported metrics often representing best-case scenarios.
- **Computational Requirements:** Complex DL models demand significant computational resources, raising questions about practical scalability.
- **Overfitting Risks:** High reported accuracies may reflect overfitting to specific datasets rather than genuine predictive power.
- **Limited Out-of-Sample Validation:** Many studies lack rigorous testing on truly unseen market conditions or extended time periods.

3 Materials and Methods

3.1 Literature Analysis

- **Search Strategy:** Studies were sourced from Scopus, IEEE Xplore, ScienceDirect, arXiv, and Springer using keywords: “machine learning,” “deep learning,” “stock market prediction,” “LSTM,” “CNN-LSTM,” “Transformer,” and “temporal fusion.”
- **Inclusion Criteria:** Papers published between 2019 and 2026 focusing on ML/DL applications for stock prediction, reporting standardized performance metrics (RMSE, MAE, MAPE), and incorporating technical indicators were prioritized. Preference given to peer-reviewed journal articles and conference proceedings.
- **Data Extraction:** For each study, we extracted model architectures, performance metrics, technical indicator usage, dataset characteristics, and validation methodologies.

Table 1: Summary of Recent Verified Studies in Stock Market Prediction

Study	Models	Indicators	Best Metrics			Market
Joshi et al. (2025)	CNN-LSTM	Price, Volume	15% improvement	RMSE	im-	Indian NSE
Lu et al. (2020)	CNN-LSTM	OHLC, Volume	Complexity study	2020		Chinese stocks
IEEE (2025)	TFT with GAF	RSI, MACD	MAPE: 1.99%, RMSE: 5.05			Apple (NASDAQ)
Thai et al. (2026)	TFT vs LSTM	RSI, MACD, OBV	MAPE $\leq$ 2%, 40–50% MAE reduction			VIC, VRE, VHM
arXiv (2025)	LSTM	Multi-feature	MAPE: 2.72–8.05%			NASDAQ stocks
Baek (2024)	GA-CNN-LSTM	Technical indicators	Training 0.0064	RMSE:		Korean stocks
Lim et al. (2021)	TFT	Multi-variate	Multi-horizon forecasting			Various

3.2 Model Descriptions

Table 2: Characteristics of Evaluated Models

Model	Architecture	Strengths	Weaknesses
SVM	Hyperplane optimization	Robust to noise, effective with small datasets	Limited temporal modeling, requires feature engineering
Random Forests	Decision tree ensemble	Non-linear patterns, feature importance	Poor sequential relationships, static assumptions
XGBoost	Gradient-boosted trees	Automated feature selection, handles missing data	Limited temporal modeling without engineering
ARIMA	Statistical time series	Interpretable, short-term accuracy	Linear assumptions, volatility sensitivity
LSTM	Recurrent neural network	Long-term dependencies, sequential learning	High data requirements, overfitting risk
CNN	Convolutional layers	Spatial feature extraction	Limited sequence modeling
CNN-LSTM	Hybrid architecture	Combines spatial and temporal learning	Higher computational cost
TFT	Attention-based	Multi-horizon, interpretable attention	Significant computational resources

3.2.1 Long Short-Term Memory (LSTM)

Overview: LSTM networks are specifically designed for sequential data, capturing long-term dependencies through gated memory cells that selectively retain or discard information.

Architecture: LSTM units feature memory cells with forget, input, and output gates. Hyperparameters include unit count (64–256), layers (1–4), and dropout rate (0.2–0.5).

Formula:

- Forget gate: $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$
- Input gate: $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$
- Cell state: $C_t = f_t \cdot C_{t-1} + i_t \cdot \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$
- Output gate: $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$
- Hidden state: $h_t = o_t \cdot \tanh(C_t)$

Stock Prediction Performance: Recent studies report MAPE ranging from 2.72% to over 16% depending on stock characteristics and market conditions. Single-layer LSTM achieved MAE of 369.32 and RMSE of 412.84 on S&P 500 data, representing 92.46% accuracy [7]. Performance is highly dependent on training data quality and hyperparameter tuning.

3.2.2 CNN-LSTM

Overview: CNN-LSTM integrates CNN’s spatial feature extraction with LSTM’s temporal modeling, creating an architecture optimized for financial time series with technical indicators.

Architecture: CNN layers first extract features from input sequences (e.g., technical indicator matrices), feeding these to LSTM units for temporal processing. Final dense layers produce predictions.

Stock Prediction Performance: Joshi et al. (2025) reported 15% RMSE improvement over standalone CNN and LSTM models on Indian NSE stocks [6]. Lu et al. (2020) demonstrated effective forecasting across 7,127 trading days combining OHLC data with technical features [2]. Genetic algorithm-optimized versions achieved training RMSE of 0.0064 [4].

3.2.3 Temporal Fusion Transformers (TFT)

Overview: TFT utilizes self-attention mechanisms combined with recurrent processing to model long-range dependencies across multiple time horizons, with interpretable attention weights.

Architecture: Multi-head self-attention layers process temporal inputs with positional encodings. The model incorporates static covariates (e.g., stock metadata) with time-varying features. Gated residual networks enhance feature selection.

Formula:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

Stock Prediction Performance:

- IEEE (2025): MAPE of 1.99%, RMSE of 5.05 on Apple stock with GAF image features [9]
- Thai et al. (2026): 40–50% MAE reduction vs LSTM/BiLSTM, MAPE ↓ 2% on Vietnamese stocks [10]

- Lim et al. (2021): Superior multi-horizon forecasting with interpretable attention mechanisms [3]

However, TFT requires significant computational resources and careful hyperparameter tuning to achieve these results.

4 Results and Discussion

4.1 Performance Synthesis

Based on recent peer-reviewed literature, model performances vary significantly by task, dataset, and market conditions:

- **Traditional ML (SVM, Random Forests, XGBoost):** Directional prediction accuracy typically 55–65% in realistic settings, though some studies report higher values. Suitable for feature-rich environments with limited temporal dependencies.
- **ARIMA:** Short-term accuracy approximately 89.8%, declining significantly over longer horizons. Best suited for stable, low-volatility periods.
- **LSTM:** MAPE ranging from 2.72% to 16.21% depending on stock and market conditions. Training RMSE values from 207.34 to 614 reported across studies. Particularly effective when sufficient historical data available.
- **CNN-LSTM:** Demonstrates 15% RMSE improvement over standalone models in controlled studies. Training RMSE as low as 0.0064 reported with genetic algorithm optimization, though generalization remains uncertain.
- **Temporal Fusion Transformers:** Best-case MAPE of 1.99–2% reported on specific stocks. Represents current state-of-the-art for multi-horizon forecasting but requires substantial computational investment.

Table 3: Representative Performance Metrics from Recent Studies

Stock/Index	Model	MAE	RMSE	MAPE (%)
S&P 500	LSTM (favorable)	369.32	412.84	7.54
S&P 500	ARIMA	462.1	614	10.2
Apple (NASDAQ)	LSTM	–	–	2.72
Apple (NASDAQ)	TFT with GAF	–	5.05	1.99
AAL	Transformer	0.0793	0.0923	8.05
VIC (Vietnam)	TFT	40–50% better than LSTM, MAPE ↓ 2%		
Indian NSE	CNN-LSTM	15% RMSE improvement over standalone		

4.2 Technical Indicators

Technical indicators significantly enhance model performance across architectures:

- **Usage Patterns:** RSI, MACD, and Moving Averages incorporated in most high-performing models

- **Performance Impact:** Studies show improvements ranging from 8–15% when incorporating well-chosen technical indicators
- **Optimal Combinations:** Multi-indicator approaches (e.g., RSI + MACD + OBV) consistently outperform single-indicator models

4.3 Critical Discussion

Performance Context: The reported performance metrics, while impressive, must be interpreted carefully:

- Best results often represent optimal conditions on specific datasets
- Performance varies significantly across different stocks and time periods
- Short evaluation periods may not reflect model robustness during market regime changes
- High reported accuracies may indicate overfitting rather than genuine predictive power

Practical Considerations:

- **Computational Costs:** Transformers and complex hybrid models require substantial computational resources (GPUs, extended training time)
- **Data Requirements:** Deep learning models need extensive historical data (often 5–10+ years) for effective training
- **Interpretability:** Traditional ML models offer better interpretability, important for risk management and regulatory compliance
- **Real-time Deployment:** Model inference latency can impact practical trading applications

Model Selection Guidelines:

1. For resource-constrained environments: XGBoost or Random Forests with engineered features
2. For moderate complexity with good performance: Standalone LSTM with technical indicators
3. For state-of-the-art performance with computational resources: CNN-LSTM or TFT
4. For interpretability requirements: Traditional ML or TFT (with attention visualization)

4.4 Limitations

Several limitations should be acknowledged:

- Model performances extracted from diverse studies with varying methodologies and evaluation periods
- Focus primarily on technical indicators and price-based features, with limited exploration of alternative data sources
- Inherent non-stationarity of financial markets means historical performance may not generalize to future conditions
- Publication bias may favor studies reporting higher performance metrics

5 Conclusions

This comprehensive review of recent peer-reviewed literature identifies the following key findings:

- 1) **Deep Learning Superiority:** LSTM, CNN-LSTM, and Transformer architectures generally outperform traditional ML approaches for temporal stock prediction, though at increased computational cost.
- 2) **Hybrid Architectures Show Promise:** CNN-LSTM models demonstrate 15% RMSE improvements over standalone approaches, effectively combining spatial and temporal feature extraction.
- 3) **Transformers Represent Current State-of-the-Art:** Temporal Fusion Transformers achieve MAPE below 2% in favorable conditions, with the added benefit of interpretable attention mechanisms.
- 4) **Performance is Context-Dependent:** Model effectiveness varies significantly across stocks, time periods, and market conditions. Reported metrics often represent best-case scenarios.
- 5) **Practical Trade-offs Matter:** Model selection should balance predictive performance, computational requirements, interpretability needs, and deployment constraints.

Recommendations for Practitioners:

- Start with simpler models (LSTM, XGBoost) before investing in complex architectures
- Conduct rigorous out-of-sample testing across multiple time periods and market conditions
- Incorporate multiple technical indicators but beware of multicollinearity
- Consider ensemble approaches combining complementary model strengths
- Maintain realistic expectations about prediction accuracy given market complexity

Future Research Directions:

- Systematic evaluation of model robustness across diverse market regimes
- Integration of alternative data sources (sentiment, macroeconomic factors, order flow)
- Development of more interpretable deep learning architectures
- Investigation of transfer learning approaches across different markets
- Enhanced uncertainty quantification in predictions

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Conflict of Interest

The author declares no conflict of interest.

Data Availability

This review synthesizes findings from published studies cited in the references. No primary data were collected for this analysis.

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